

Longest Substring Without Repeating Characters

1. Objective

To develop a Java program to find the **longest substring without repeating characters** from a given input string and display the substring along with its length.

2. Algorithm

1. Start the program.
2. Import the Scanner class.
3. Read a string from the user.
4. Initialize an empty string longest to store the longest substring.
5. Use an outer loop to select each character as the starting position.
6. For every starting position, initialize an empty string current.
7. Use an inner loop to select characters continuously from the starting position.
8. Check whether the selected character is already present in current.
9. If the character is repeated, stop the inner loop.
10. Otherwise, add the character to current.
11. Compare the length of current with longest.
12. If current is longer, store it in longest.
13. Display the longest substring and its length.
14. Stop the program.

3. UML Class Diagram

LongestSubstring + main(String args[]) : void
Uses
Scanner + nextLine() + close()

4. Source Code

```
import java.util.Scanner;

class LongestSubstring{

    public static void main(String args[]){

        Scanner sc = new Scanner(System.in);

        // Read the input string

        System.out.print("Enter a string: ");

        String str = sc.nextLine();

        // Store the longest substring

        String longest = "";

        // Select the starting position

        for(int i = 0; i < str.length(); i++){

            // Store the current substring

            String current = "";

            // Select characters continuously from position i

            for(int j = i; j < str.length(); j++){

                char ch = str.charAt(j);

                // Check whether the character is already present

                if(current.indexOf(ch) != -1){

                    // Repeated character found

                    break;

                }

                // Add the character to current substring

                current = current + ch;

                // Check whether current substring is longer

                if(current.length() > longest.length()){

                    longest = current;

                }

            }

        }

    }

}
```

```
        }  
    }  
}  
  
// Display the longest substring and its length  
System.out.println("String = " + longest);  
System.out.println("Length = " + longest.length());  
sc.close();  
}  
}
```

5. Output

Sample Input:

Enter a string: abcabcbb

Output:

String = abc

Length = 3

Another example:

Sample Input:

Enter a string: pwwkew

Output:

String = wke

Length = 3